



**BIT2083 FUNDAMENTAL OF COMPUTATIONAL THINKING:
PYTHON**

**FINAL PROJECT - GROUP
(40%)**

ALL SECTIONS

**FACULTY OF INFORMATION TECHNOLOGY
CITY UNIVERSITY MALAYSIA
CYBERJAYA CAMPUS**

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INSTRUCTIONS TO STUDENTS

1. This **project brings 40%** from overall course assessment.
2. You are allowed to refer to the learning material to produce a quality assignment output within a specified time.
3. It is a **GROUP assignment of 5 students maximum** and **full marks is 40%**.
4. Complete all questions given.
5. Only a single representative or team leader can submit the assignment.
6. The submission of the assignment must include you & your members'
 - a. **FULL NAME,**
 - b. **STUDENT ID,**
 - c. **CLASS CODE,**
 - d. **PROGRAM,**
 - e. **NRIC/PASSPORT NUMBER.**
7. Please upload the completed assignment via your CityU LMS.
8. All submission formats are soft-copy/digital output only.
9. **Presentation day is on TBA.**

FINAL PROJECT RUBRIC (40%)

Section	Description	Weight	Very Poor (1)	Poor (2)	Fair (3)	Good (4)	Excellent (5)
Problem Identification	Clearly explains the real-world problem and project objective	10%	Problem unclear	Limited explanation	Adequate explanation	Clear explanation	Excellent problem justification with real-world relevance
Computational Thinking	Demonstrates decomposition, abstraction, pattern recognition and algorithmic thinking	15%	No evidence	Minimal evidence	Partial application	Good application	Excellent application throughout the project
Program Design	Flowchart, algorithm or pseudocode organization	10%	Missing	Poor structure	Acceptable	Well organized	Professional and logical design
Functions & Modular Programming	Proper use of user-defined functions and modular structure	15%	No functions	Few functions	Basic modularity	Good modularity	Highly organized reusable functions
Python Programming Implementation	Correct use of variables, input/output, operators, conditionals and loops	25%	Numerous errors	Many errors	Mostly functional	Functional with minor issues	Fully functional and efficient
User Experience	Menu design, prompts, readability and usability	10%	Difficult to use	Poor interface	Acceptable	User friendly	Excellent interface and navigation

Code Quality	Comments, indentation, naming conventions and readability	10%	Poor quality	Limited readability	Acceptable	Well organized	Professional coding standards
Demonstration & Presentation	System demonstration, explanation and ability to answer questions	5%	Unable to explain	Limited understanding	Basic explanation	Good explanation	Excellent demonstration and confident explanation

SUSTAINABLE SOLUTIONS THROUGH COMPUTATIONAL THINKING

Sustainability has become one of today's greatest global challenges. Communities around the world face issues such as food waste, excessive energy consumption, inefficient resource management, and increasing environmental impacts caused by human activities. Although these challenges are global in nature, many can be addressed through simple technological solutions that encourage better decision-making and responsible resource usage.

In this project, students will apply Computational Thinking concepts and Python programming to develop a console-based application that addresses a real-world sustainability problem aligned with one of the selected Sustainable Development Goals (SDGs). The application should assist users in making informed decisions, performing useful calculations, or managing information that contributes to a more sustainable lifestyle or community.

Students are expected to demonstrate the use of decomposition, pattern recognition, abstraction, algorithm design, and problem-solving techniques while developing a functional Python application using the programming concepts learned throughout the semester.

TASK DESCRIPTION

Each group is required to select **ONE (1)** of the following Sustainable Development Goals (SDGs):

1. [SDG 2: Zero Hunger](#)
2. [SDG 7: Affordable and Clean Energy](#)
3. [SDG 11: Sustainable Cities and Communities](#)
4. [SDG 13: Climate Action](#)

Based on the selected SDG, identify a real-world problem that can be solved through a simple Python console application. The proposed solution should demonstrate computational thinking principles and utilise Python programming concepts learned in this course. The selected SDG(s) should be clearly identified and justified in the project report.

The main task is to:

1. Identify a real-world sustainability problem.
2. Develop a Python console application that provides a practical solution to the identified problem.
3. Explain how the application contributes to the selected SDG(s).

The application must demonstrate computational thinking concepts and apply Python programming techniques learned throughout the course.

Students may propose their own project idea or select from the examples below.

SDG 2: Zero Hunger

- Food Donation Management System
- Household Food Waste Tracker
- School Meal Planning Calculator
- Food Inventory Expiry Checker
- Community Food Distribution Planner

SDG 7: Affordable and Clean Energy

- Household Electricity Consumption Calculator
- Solar Energy Savings Estimator
- Home Energy Usage Tracker
- Monthly Electricity Bill Estimator
- Energy-Efficient Appliance Comparison Tool

SDG 11: Sustainable Cities and Communities

- Public Transportation Fare Calculator
- Smart Parking Fee Calculator
- Household Recycling Tracker
- Community Event Registration System
- Waste Collection Schedule Planner

SDG 13: Climate Action

- Personal Carbon Footprint Calculator
- Carbon Emission Estimator for Transportation
- Tree Planting Impact Calculator
- Household Water Consumption Tracker
- Eco-Friendly Lifestyle Assessment System

PROJECT SCOPE & APPLICATION REQUIREMENTS

The application must be developed using Python programming language and demonstrate the computational thinking concepts learned throughout the course.

Requirement	Description
Programming Language	Python
Application Type	Console-based application
Menu System	Menu-driven interface using loops
User Input	Accept user inputs using <code>input()</code>
Output	Display meaningful and formatted outputs
Variables	Appropriate use of variables and data types
Operators	Mathematical and logical operators
Functions	Minimum five (5) user-defined functions
Conditional Statements	Appropriate use of <code>if</code> , <code>elif</code> , and <code>else</code>
Loops	Appropriate use of <code>for</code> and/or <code>while</code> loops
Problem Solving	Demonstrate computational thinking in solving the selected problem
Error Handling	Basic input validation for invalid user inputs
Comments	Appropriate inline comments and documentation
Coding Style	Proper indentation and meaningful variable/function names

FUNCTIONAL REQUIREMENTS (Minimum)

The application shall include **at least five (5) major functions** appropriate to the selected problem. Typical functions may include:

1. Display Main Menu
2. Enter or Record Information
3. Perform Calculations or Analysis
4. Generate and Display Results
5. Exit the Application

Additional functions may be implemented depending on the project requirements.

PROJECT SUBMISSION

1. Python Application (Source Code)

1.1. GitHub repository (put this url at your report)

- Well-structured packages
- Meaningful class and method names
- Proper comments

2. Project Report & Slide (Soft-copy PDF)

The report & slide must include the following:

- 2.1. Introduction & SDG Background
- 2.2. Problem Statement
- 2.3. System Objectives
- 2.4. Application Design (UML Class Diagram)
- 2.5. Implementation Details
- 2.6. Testing & Sample Outputs
- 2.7. Discussion & Limitations
- 2.8. Conclusion

3. Presentation & Demonstration (Working Application)

- 3.1. Prepare slide presentation
- 3.2. Present the project for about 15 minutes per group
- 3.3. Live system demo
- 3.4. All members must participate

ASSIGNMENT END
